

Harikaran C

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EDUCATION

SSN College of Engineering

B.Tech. Information Technology, CGPA: 8.50 / 10

Chennai, India

Aug. 2023 – May 2027

Alwin International Public School (CBSE)

10th Grade: 474 / 500, 12th Grade: 482 / 500, JEE Mains (April 2023): 96 Percentile

Chennai, India

June 2019 – Aug. 2022

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL

Frameworks/Technologies: Flask, Node.js, Express, React, MySQL, MongoDB, REST APIs, JWT, WebSockets, BullMQ, OAuth, Flutter(Basics)

Tools: Git, Github, Postman

CS Fundamentals: Data Structures Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems (OS), Computer Networks, Artificial Intelligence Machine Learning (AI/ML)

EXPERIENCE

Software Developer Intern

Adamsbridge Services Private Limited

May 2026 – June 2026

Chennai, Tamil Nadu

- Developed backend services for an AI-powered email copilot, integrating Gmail APIs, OAuth 2.0, and LLMs to automate email understanding and organization
- Designed asynchronous processing workflows using Redis, BullMQ, background workers, and scheduled jobs to support scalable email ingestion and analysis
- Built intelligent productivity features including email prioritization, task/deadline extraction, notifications, and automated daily digests

PROJECTS

AI-Powered Email Copilot | *MERN, Gemini API, OAuth, BullMQ, node-cron*

May 2026 – June 2026

- Engineered a scalable event-driven architecture using Redis, BullMQ, retries, exponential backoff, and background workers for reliable email processing
- Built semantic inbox search and conversational email assistance using embeddings, MongoDB Atlas Vector Search, Retrieval-Augmented Generation (RAG), and multi-turn chat memory
- Developed a multi-LLM workflow with Groq and Gemini integration for email summarization, classification, task extraction, and context-aware question answering

Smart Ambulance Dispatch System | *Flask, React, MongoDB, WebSockets, TomTom API* Jan. 2025 – Apr. 2025

- Built and deployed a full-stack emergency response platform for real-time ambulance dispatch automation
- Implemented event-driven WebSocket workflows for live ambulance tracking, driver standby management, emergency task assignment, and real-time patient notifications
- Developed a heuristic-based route optimization engine that continuously evaluated traffic, distance, and travel-time metrics to select the most efficient route from multiple TomTom alternatives
- Designed an operations dashboard for monitoring fleet availability, hospital capacity, dispatch history, and response-time analytics

Multimodal Sleep Stage Classification | *Python, TensorFlow/Keras, MNE, Scikit-learn* Sept. 2025 – Apr. 2026

- Developed an end-to-end multimodal sleep stage classification pipeline using EEG, EOG, and EMG signals from the PhysioNet Sleep-EDF dataset
- Designed a multi-branch CNN-LSTM architecture with channel attention and cross-modal attention mechanisms to learn spatial and temporal sleep patterns
- Implemented subject-wise training and evaluation using GroupShuffleSplit, signal preprocessing with MNE, and class imbalance handling to improve model generalization
- Evaluated performance using accuracy, Cohen's Kappa, confusion matrices, and per-class metrics on unseen subjects

ACHIEVEMENTS AND CERTIFICATIONS

Competitive Programming – Solved 800+ problems over Codeforces, Leetcode

Codeforces Max Rating: 1450, Specialist

NPTEL Elite Silver – Programming in Java – IIT Kharagpur (12-week)

NPTEL Elite Silver – Getting Started with Competitive Programming – IIT GN (12-week)

LEADERSHIP

SSN Coding Club | *Deputy Lead, CP Vertical*

Aug. 2025 – Present

- Organized coding competitions and mentored juniors, enhancing problem-solving culture within the club

SSN Photography Club | *Senior Core*

Jan. 2024 – Present

- Coordinated events and workshops while mentoring peers to enhance creative collaboration and technical skills